

Akshar Patel

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SUMMARY

Ambitious and results-driven Data Scientist with robust experience in machine learning, deep learning, and data analysis. Proficient in developing and deploying data models and AI solutions, with a strong background in programming, statistical analysis, and real-world problem-solving.

EDUCATION

Charotar University of Science and Technology, Anand, Gujarat
Bachelor of Technology (B.Tech) in Information Technology
October 2021 – April 2025
CGPA: 8.7

SKILLS

Machine Learning & AI: Supervised/Unsupervised Learning, Transfer Learning, Reinforcement Learning, Ensemble Methods, Neural Networks, LSTM, CNN, NLP (Text Classification, Named Entity Recognition, Sentiment Analysis, Computer Vision).
GenAI Models: Transformers, LLMs, GANs, VAEs, NVLM, Llama, Diffusion Models.
Data Science & Analysis: Data Preprocessing, Feature Engineering, Statistical Analysis, A/B Testing, Hypothesis Testing, Fine tuning, Data Visualization (Matplotlib, Seaborn, Plotly, PowerBI, MATLAB)
Programming Languages: Python, SQL, Java, C++, C
Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-Learn, Pandas, NumPy, Hugging Face
Version Control & Deployment: Git, GitHub, Docker, Kubernetes, CI/CD Pipelines
Databases: Oracle, MySQL, MongoDB
Web Development: HTML, CSS, JavaScript, Flask, Streamlit, MERN Stack (MongoDB, Express.js, React, Node.js)
BI & Visualization Tools: PowerBI, Microsoft Excel, Word, PowerPoint.
Additional Technical Skills: Networking, Linux, Cryptography, Design and Analysis of Algorithms (DAA), Data Structures and Algorithms (DSA), Statistics, Flutter
Soft Skills: Interpersonal Communication, Time Management, Problem Solving, Project Management, Critical Thinking

PROJECTS

Chatting with Multiple PDFs using LangChain (June 2024)

- Developed a system for loading and retrieving information from multiple PDF files using LangChain and OpenAI models like ChatGPT.
- Implemented a process in Python to convert PDF files into embeddings based on selected large language models for efficient data extraction.
- Technologies Used: Python, LangChain, OpenAI API, PDF Processing Libraries*

Diabetic Retinopathy Blindness Detection (February 2024)

- Created and deployed a web application for Diabetic Retinopathy detection using deep learning and Python.
- Trained a model to analyze retinal images, achieving accurate identification of DR stages, aiding in early detection and patient care.
- Technologies Used: Python, TensorFlow, Keras, Flask, OpenCV*

Disease Detector: Multiple Disease Prediction WebApp (November 2023)

- Developed a Flask-based web application in Python for predicting diseases such as diabetes and heart disease based on input symptoms.
- Incorporated image classification for brain tumors and pneumonia, enabling users to upload X-ray images for prediction.
- Technologies Used: Python, Flask, Scikit-Learn, TensorFlow, OpenCV*

DocsCat: Document Scanner and Translate (April 2023)

- Built an NLP-based application in Python for processing and translating text extracted from images.
- Utilized natural language processing techniques to scan images and translate the text into various languages.
- Technologies Used: Python, NLTK, SpaCy, Tesseract OCR, Google Translate API*

WORK EXPERIENCE

Business Optima, California, United States (Remote)

AI Developer (October 2024 – Present)

- Designed and developed demos for cutting-edge Generative AI models, including Transformers, GANs, VAEs, Diffusion Models, LLMs, Retrieval-Augmented Generation (RAG), and Multi-Agent AI systems. These demos showcased the potential of AI in solving real-world challenges across various domains.
- Implemented end-to-end automation workflows using tools such as PhantomBuster, Zapier, Twilio, Brevo, and Zendesk. Streamlined processes for lead generation, ticketing systems, and communication pipelines, enhancing operational efficiency and user experience.
- *Technologies Used: Transformers, LLMs, RAG, Multi-Agent AI, Automation tools like Zapier and PhantomBuster.*

Scale Capacity, California, United States (Remote)

Data Science Intern (May 2024 – August 2024)

- In this role, I worked extensively with cloud technologies and Python to develop data-driven solutions.
- I applied my knowledge to analyze complex datasets, automate processes, and optimize workflows, significantly improving the efficiency of their operations.
- I worked closely with the company's AI/ML team to solve real-world problems.
- *Technologies Used: Python, TensorFlow, Keras, Scikit-Learn, AWS*

Helios Solutions, Vadodara, Gujarat (On-Site)

Data Analyst Intern (May 2023 - June 2023)

- Created models to meet the company's specific needs while maintaining high accuracy.
- Coordinated the development of VisuVerse, integrating data analysis with visualization for interactive online insights.
- Developed an interactive dashboard that reduced data analysis time by 30%.
- *Technologies Used: Python, Streamlit, Tensorflow, Scikit-Learn*

Suvidha Foundation, Mumbai, Maharashtra (Remote)

Machine Learning Intern (Feb 2023 - April 2023)

- Building predictive models and algorithms to solve complex business problems
- Preprocessing and cleaning data to ensure its quality and integrity. Developed and implemented various machine learning models to solve complex problems and analyze large datasets effectively.
- *Technologies Used: Python, Scikit-Learn, Pandas, TensorFlow, Keras*

CERTIFICATIONS

Machine Learning Specialization, Stanford, Coursera

- Developed skills in building predictive models using supervised and unsupervised learning for effective problem-solving and decision-making.

AWS Machine Learning Path, AWS

- Acquired expertise in building, training, and deploying ML models using AWS services like SageMaker, S3, and Lambda, with a focus on scalable solutions and AI concepts.

Deep Learning Specialization, Stanford, Coursera

- Gained proficiency in sequence models, neural networks, and convolutional networks, applied to computer vision, NLP, and sequence generation.

GenAI using LLMs, Stanford, Coursera

- Mastered advanced AI techniques, including generative modeling and reinforcement learning, enhancing knowledge of AI technologies.
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RESEARCH AND PUBLICATION

Enhanced Brain Tumor Classification Using Transfer Learning: A Comparative Analysis of Pre-trained CNN Models on MRI Images

International Conference on Data Science and Analytics (ICDSA), 2024

Conducted a comparative study on the effectiveness of various pre-trained CNN models for classifying brain tumors using MRI images. Demonstrated significant improvements in classification accuracy through transfer learning techniques.

Fine-tuning Deep Learning Model Using Transfer Learning for Pneumonia Diagnosis

International Conference on Technology for Sustainable Development (ICT4SD), 2024

Explored the application of transfer learning to fine-tune deep learning models for pneumonia diagnosis. Achieved enhanced diagnostic performance and contributed to advancements in automated medical imaging analysis.

ACHIEVEMENTS

Maker 's Fest Vadodara 2024 - Silver in Technology Category

Project: Diabetic Retinopathy Blindness Detection using Retinal Images - AI

NPTEL Topper – Silver in Fundamentals of Artificial Intelligence

Description: Gained a deep understanding of key AI concepts, including machine learning, neural networks, and problem-solving strategies